# MSG No. 49.268 $Pccm'$ [ Type III, orthorhombic ]

Table 1: Wyckoff site: 2a, site symmetry: ..2/m'

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, 0]$ | [1,2,7,8] |
| 2 | $\left[0, 0, \frac{1}{2}\right]$ | [3,4,5,6] |

Table 2: Wyckoff site: 2b, site symmetry: ..2/m'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | [1,2,7,8] |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | [3,4,5,6] |

Table 3: Wyckoff site: 2c, site symmetry: ..2/m'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, 0\right]$ | [1,2,7,8] |
| 2 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | [3,4,5,6] |

Table 4: Wyckoff site: 2d, site symmetry: ..2/m'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, 0, 0\right]$ | [1,2,7,8] |
| 2 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | [3,4,5,6] |

Table 5: Wyckoff site: 2e, site symmetry: 2'2'2

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, 0, \frac{1}{4}\right]$ | [1,2,5,6] |
| 2 | $\left[0, 0, \frac{3}{4}\right]$ | [3,4,7,8] |

Table 6: Wyckoff site: 2f, site symmetry: 2'2'2

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, 0, \frac{1}{4}\right]$ | [1,2,5,6] |
| 2 | $\left[\frac{1}{2}, 0, \frac{3}{4}\right]$ | [3,4,7,8] |

Table 7: Wyckoff site: 2g, site symmetry: 2'2'2

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, \frac{1}{4}\right]$ | [1,2,5,6] |

| | | |
|---|---|---|
| 2 | $\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$ | `[3,4,7,8]` |

Table 8: Wyckoff site: `2h`, site symmetry: `2'2'2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$ | `[1,2,5,6]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$ | `[3,4,7,8]` |

Table 9: Wyckoff site: `4i`, site symmetry: `2'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ 0,\ \frac{1}{4}\right]$ | `[1,5]` |
| 2 | $\left[-x,\ 0,\ \frac{1}{4}\right]$ | `[2,6]` |
| 3 | $\left[-x,\ 0,\ \frac{3}{4}\right]$ | `[3,7]` |
| 4 | $\left[x,\ 0,\ \frac{3}{4}\right]$ | `[4,8]` |

Table 10: Wyckoff site: `4j`, site symmetry: `2'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{1}{2},\ \frac{1}{4}\right]$ | `[1,5]` |
| 2 | $\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$ | `[2,6]` |
| 3 | $\left[-x,\ \frac{1}{2},\ \frac{3}{4}\right]$ | `[3,7]` |
| 4 | $\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$ | `[4,8]` |

Table 11: Wyckoff site: `4k`, site symmetry: `.2'.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ y,\ \frac{1}{4}\right]$ | `[1,6]` |
| 2 | $\left[0,\ -y,\ \frac{1}{4}\right]$ | `[2,5]` |
| 3 | $\left[0,\ y,\ \frac{3}{4}\right]$ | `[3,8]` |
| 4 | $\left[0,\ -y,\ \frac{3}{4}\right]$ | `[4,7]` |

Table 12: Wyckoff site: `4l`, site symmetry: `.2'.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\ y,\ \frac{1}{4}\right]$ | `[1,6]` |
| 2 | $\left[\frac{1}{2},\ -y,\ \frac{1}{4}\right]$ | `[2,5]` |
| 3 | $\left[\frac{1}{2},\ y,\ \frac{3}{4}\right]$ | `[3,8]` |
| 4 | $\left[\frac{1}{2},\ -y,\ \frac{3}{4}\right]$ | `[4,7]` |

Table 13: Wyckoff site: `4m`, site symmetry: `..2`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | `[1,2]` |
| 2 | $\left[0,\ 0,\ z+\frac{1}{2}\right]$ | `[3,4]` |
| 3 | $\left[0,\ 0,\ \frac{1}{2}-z\right]$ | `[5,6]` |
| 4 | $[0,\ 0,\ -z]$ | `[7,8]` |

Table 14: Wyckoff site: `4n`, site symmetry: `..2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$ | `[1,2]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[3,4]` |
| 3 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | `[5,6]` |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$ | `[7,8]` |

Table 15: Wyckoff site: `4o`, site symmetry: `..2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ z\right]$ | `[1,2]` |
| 2 | $\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[3,4]` |
| 3 | $\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | `[5,6]` |
| 4 | $\left[0,\ \frac{1}{2},\ -z\right]$ | `[7,8]` |

Table 16: Wyckoff site: `4p`, site symmetry: `..2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\ 0,\ z\right]$ | `[1,2]` |
| 2 | $\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$ | `[3,4]` |
| 3 | $\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$ | `[5,6]` |
| 4 | $\left[\frac{1}{2},\ 0,\ -z\right]$ | `[7,8]` |

Table 17: Wyckoff site: `4q`, site symmetry: `..m'`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ 0]$ | `[1,8]` |
| 2 | $[-x,\ -y,\ 0]$ | `[2,7]` |
| 3 | $\left[-x,\ y,\ \frac{1}{2}\right]$ | `[3,6]` |
| 4 | $\left[x,\ -y,\ \frac{1}{2}\right]$ | `[4,5]` |

Table 18: Wyckoff site: `8r`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | `[1]` |

*continued ...*

Table 18

| No. | position | mapping |
|---|---|---|
| 2 | $[-x, -y, z]$ | [2] |
| 3 | $\left[-x, y, z+\frac{1}{2}\right]$ | [3] |
| 4 | $\left[x, -y, z+\frac{1}{2}\right]$ | [4] |
| 5 | $\left[x, -y, \frac{1}{2}-z\right]$ | [5] |
| 6 | $\left[-x, y, \frac{1}{2}-z\right]$ | [6] |
| 7 | $[-x, -y, -z]$ | [7] |
| 8 | $[x, y, -z]$ | [8] |